Capturing the Dynamics of Implied Volatility Surfaces: A Deep Learning Approach for NIFTY50 Index Options

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by

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Abstract

This study presents a deep learning approach to forecasting the dynamics of implied volatility surfaces (IVS) for NIFTY50 index options. The primary objective is to develop a comprehensive model that captures both the term structure and volatility skew across different strike prices and expiration dates. Implied volatilities are computed using the Black-Scholes formula, and the study evaluates traditional econometric models such as Vector Auto Regression (VAR) and Vector Error Correction Model (VECM) alongside advanced deep learning models such as Long Short-Term Memory (LSTM) and Attention-based LSTM (Att-LSTM). The models are trained on minute-level NIFTY50 option data, and their performance is compared across various forecasting horizons. The results demonstrate that the Att-LSTM model consistently outperforms both traditional and LSTM models, with superior accuracy in capturing the dynamics of the volatility surface. The findings highlight the superiority of deep learning for implied volatility forecasting, ultimately aiding in improved risk management and option pricing in financial markets.

Table of Contents

List of Figures

List of Tables

Main Content

1 Introduction

Volatility is a fundamental concept in financial markets, referring to the extent of fluctuation or uncertainty in the return of a financial asset. It is a measure of the asset's risk and plays a crucial role in various aspects of finance, including asset pricing, risk management, and derivative valuation. Understanding and accurately forecasting volatility is of utmost importance for market participants, as it allows market participants to make more informed trading and risk management decisions regarding options contracts, optimize their portfolios, and manage their risk exposure effectively.

The two primary types of volatility in the financial world are: historical (realized) volatility and implied volatility. Historical volatility is based on the past performance of an asset and can be calculated using the asset's historical price data. It provides a measure of the asset's actual price fluctuations over a certain period. On contrary, implied volatility is a forward-looking indicator that captures market expectations for future price fluctuations. Unlike historical volatility, implied volatility cannot be directly observed and must be calculated from derivative prices such as options, using a pricing model like the Black-Scholes formula.

Implied volatility is crucial as it reflects the market's aggregate outlook on future volatility, incorporating factors like market expectations, economic conditions, and supply-demand forces. It is a key input in option pricing models and is used to determine the fair value of options. Fluctuations in implied volatility can substantially impact option prices and broader market sentiment. Therefore, the ability to accurately forecast implied volatility is crucial for market participants, especially those involved in derivative trading.

Forecasting implied volatility is a challenging task due to its dynamic and market-driven nature. Accurate forecasts of implied volatility enable institutions to assess their investment risk, make informed trading decisions, and optimize their portfolios. It helps in pricing options more precisely, developing effective hedging strategies, and identifying potential trading opportunities. Implied volatility predictions offer valuable insights into market sentiment and serve as a risk management tool.

Given the significance of implied volatility in financial markets, extensive research has been conducted to develop models and techniques for forecasting implied volatility. These models range from traditional econometric approaches to more advanced machine learning and deep learning methods. The goal is to capture the complex dynamics and patterns in implied volatility and provide accurate and reliable forecasts to support decision-making in various financial applications. [1]

[1] The code used for this work can be found at: https://github.com/TomChow01/volatility-surface.git

1.1 Problem Statement

The traditional Black-Scholes model assumes constant volatility, which is inconsistent with the observed implied volatility (IV) behavior in real-world markets. IV exhibits two key properties: *1) IV term structure*, where future IV is typically higher than today's IV, and *2) IV skew or IV smile* [Figure 1a], where IV levels vary across different option strikes due to market sentiment and investor preferences. Empirical studies have shown that IV varies with strike prices and time-to-maturities, contradicting the Black-Scholes assumptions. The implied volatility surface (IVS) [Figure 1b] captures both the IV skew and IV term structure, and short-term deviations from the classical IVS shape can be observed. Previous research has primarily focused on cross-sectional results using non-parametric methods for volatility pricing, while machine learning techniques have shown promise in three main areas: forecasting historical volatility, developing non-parametric frameworks for option pricing, and time series forecasting of IV and related products. This thesis contributes to the third area, proposing a data-driven machine learning technique for forecasting IV and generating multi-step forecasts of both IV skew and IV term structure through the IVS. The main objective is to develop an end-to-end forecasting framework for the entire volatility surface using advanced machine learning techniques, capturing the complex dynamics and patterns present in the IVS. The proposed framework will be evaluated using real-world market data, considering various market conditions and major events. The performance will be compared against traditional models and benchmarks to assess its effectiveness and robustness. The ultimate goal is to contribute to the field of implied volatility forecasting by developing a comprehensive and data-driven solution that can aid market participants to make informed choices and managing risk in the face of ever-changing market dynamics.

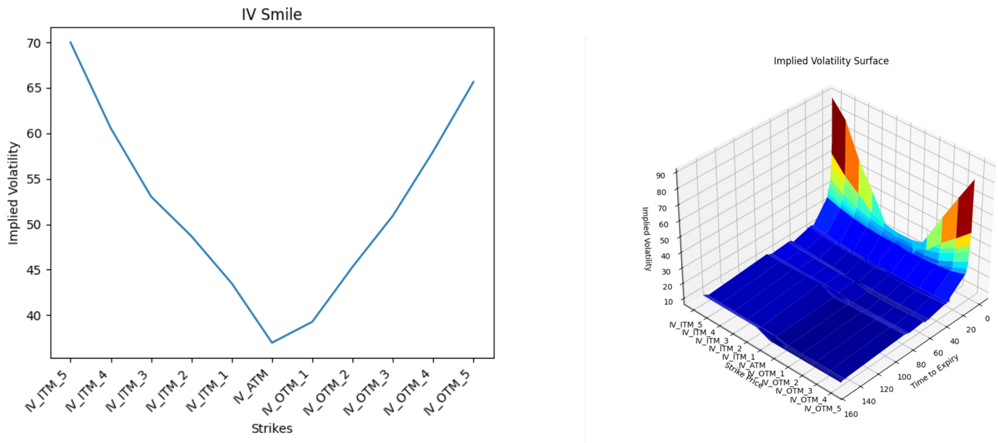
1.2 Background

1.2.1 Introduction to Financial Derivatives and Options

In the realm of finance, the term "derivative" refers to a contract whose value is based on one or more underlying assets or asset groups, commonly known as an "index" or "benchmark." The word "derivative" itself suggests that the financial instrument's worth is not intrinsic but rather depends on the value of another asset or set of assets.

Derivatives are contracts formed between two parties, which can be traded on a financial exchange or over-the-counter (OTC). The underlying assets can take various forms, such as stocks, bonds, commodities, currencies, interest rates, or even market indices. For instance, a derivative contract might be based on the price of a single stock like Apple Inc. (AAPL), or it could be tied to the performance of a stock market index like the S&P 500.

Derivatives enable investors to gain exposure to an asset without direct ownership. This characteristic allows investors to hedge risks, speculate on price changes, or leverage their positions. Derivatives also provide a way to transfer risks from one party to another, as the contract's value fluctuates with the underlying asset's price changes.



(a) Implied Volatility Smile

(b) Implied Volatility Surface

Figure 1: (a) Implied volatility smile for NIFTY index options expiring on July 13, 2023, with 24 hours remaining to expiration. The implied volatilities are plotted against option strikes (b) Implied volatility surface for NIFTY index options expiring on July 13, 2023. The surface shows implied volatilities plotted against both strikes and time to expiration.

There are several types of commonly traded derivatives, each with its own pricing formula and risk profile. Some well-known examples of derivatives include futures, forwards, options, and swaps. Futures contracts obligate the buyer to purchase, or the seller to sell, an asset at a specified future date and price. Forwards are similar to futures but offer more customization and are traded over-the-counter (OTC). Options give the buyer the right, but not the obligation, to buy (call) or sell (put) an asset at a specified price (strike price) within a certain time frame (expiration date). Swaps involve agreements to exchange cash flows or liabilities between two different financial instruments, often used to hedge against interest rate or credit risks.

Individual investors and hedge funds utilize derivatives for various purposes. For instance, an investor might use put options to safeguard their stock portfolio from potential declines. By exercising a put option, the investor can sell shares at the strike price if the stock price drops below it, thereby limiting potential losses.

Derivatives are also used extensively in risk management. For example, a company dependent on a consistent supply of a specific commodity (e.g., an airline that needs jet fuel) can utilize futures contracts to secure a fixed future price for that commodity. This strategy helps the company budget more effectively and reduces its exposure to price volatility.

Moreover, derivatives can be used for speculation. They allow traders to speculate on the price movement of an asset without needing to own it directly. This approach can lead to significant profits if the trader's prediction is correct, but it also carries substantial risks.

In contrast to traditional financial instruments like stocks and bonds, derivatives are often more complex and can be riskier. The value of a derivative is contingent on the underlying asset, and the contract itself might have intricate terms and conditions. Therefore, investors need to thoroughly understand the risks and mechanics of derivatives before entering into such contracts.

1.2.2 Options and Their Pricing

Options are financial derivatives that grant the holder the right, but not the obligation, to purchase (call option) or sell (put option) an underlying asset at a set price (strike price) by or on a certain date (expiration date). The underlying asset may include a stock, bond, commodity, currency, or index. Options provide flexibility, leverage, and the opportunity to profit in both rising and falling markets.

Key characteristics of options include:

- **Leverage**: Options enable investors to control a larger position with a smaller upfront investment, amplifying potential gains or losses.
- **Limited risk**: An option buyer's maximum loss is capped at the premium paid for the option.
- **Flexibility**: Options can be used for speculation, hedging, or income generation through various strategies.
- **Time decay**: Options decrease in value as they near expiration due to the passage of time.

Call and put options are the two basic types of option contracts. A call option gives the holder the right, but not the obligation, to buy the underlying asset at a predetermined price (the strike price) on or before the expiration date. Call options are bought when the trader expects the price of the underlying asset to rise. Conversely, a put option gives the holder the right, but not the obligation, to sell the underlying asset at the strike price on or before the expiration date. Put options are bought when the trader expects the price of the underlying asset to fall. The buyer of an option pays a premium to the seller for this right. If the price of the underlying asset moves in the desired direction, the option holder can exercise their right to buy or sell at the strike price, realizing a profit. If the price moves in the opposite direction, the option holder will let the option expire worthless, losing only the premium paid.

The Black-Scholes model, created by Fischer Black, Myron Scholes, and Robert Merton in the early 1970s, transformed the way options are priced. The model gives a mathematical framework for valuing European-style options. It is assumed that stock prices follow a log-normal distribution, there are no transaction costs or taxes, the risk-free interest rate remains constant, and arbitrage opportunities do not exist. The Black-Scholes model uses five key variables to calculate the theoretical price of an option:

- Current value of the underlying asset
- The contract's strike price
- Time to expiration
- Risk-free interest rate
- Volatility of the underlying asset

Although widely used, the Black-Scholes model has its limitations, including the assumption of constant volatility and its restriction to European-style options. To address these, other models such as the *binomial option pricing model* by Cox, Ross, and Rubinstein [1] and the *Monte-Carlo simulation* have been developed. The binomial model allows for the valuation of American-style options by modeling the price of the underlying asset in discrete time steps. The Monte-Carlo method simulates numerous potential price paths for the underlying asset to estimate the option's value.

Implied volatility is another fundamental concept in option pricing. Based on the current market price of the option, it reflects the market's forecast of the underlying asset's future volatility. Traders use implied volatility to gauge market sentiment and compare the relative expensiveness of options.

Option pricing theory continues to evolve, with new models and techniques being developed to better capture market dynamics. These advancements include stochastic volatility models, jump-diffusion models, and models incorporating market microstructure effects.

Key Option Pricing Concepts

There are several important concepts to understand when dealing with options pricing:

Strike Price: The predetermined price at which the holder of an option can buy (in the case of a call option) or sell (in the case of a put option) the underlying asset upon expiration. The strike price is fixed for the duration of the option contract.

Premium: The price paid by the option buyer to the option seller for the rights granted by the option contract. It reflects the intrinsic value plus the time value of the option.

Moneyness: Moneyness describes the relationship between the strike price of an option and the current price of the underlying asset. There are three states of moneyness:

- In-the-money (ITM): For a call option, the strike price is below the current price of the underlying asset. For a put option, the strike price is above the current price.
- At-the-money (ATM): The strike price is equal to or very close to the current price of the underlying asset.
- Out-of-the-money (OTM): For a call option, the strike price is above the current price of the underlying asset. For a put option, the strike price is below the current price.

Time-to-expiry: The remaining time until the expiration date of the option contract. All else being equal, options with more time-to-expiry have higher premiums due to greater time value. As expiration approaches, the time value decays at an accelerating rate, a phenomenon known as theta decay.

Intrinsic value: The amount by which an option is in-the-money. For call options, intrinsic value is the current stock price minus the strike price. For put options, it is the strike price minus the current stock price. Intrinsic value cannot be negative. Out-of-the-money and at-the-money options have no intrinsic value.
The choice of strike prices is a crucial aspect of options trading strategies. Traders can select from a range of available strike prices, depending on their market outlook and risk tolerance. The distribution of open interest and trading volume across different strike prices can provide insight into market sentiment and potential future price movements of the underlying asset.

These concepts are fundamental to understanding how options are priced. The Black-Scholes (BS) model and other option pricing models incorporate these factors, along with volatility and the risk-free interest rate, to determine the theoretical fair value of an option. Market forces of supply and demand then drive actual option premiums to fluctuate around these theoretical values.

2 Literature Review

In this section, I will discuss a brief history of option pricing models and the key concepts of geometric Brownian motion (GBM), which lies at the heart of most well-known option pricing models. The section will highlight the foundational option pricing model of Black-Scholes, as well as some other popular pricing models for derivatives in the stock market. Additionally, I will talk about some traditional approaches to Implied Volatility (IV) forecasting. Lastly, I will review some of the more recent methods of implied volatility forecasting using various machine learning and deep learning techniques, and how they motivated my work.

Implied Volatility

Although the concept of options dates back to the 14th century, with insurance contracts related to shipments in the Mediterranean, up until the 1960s they were viewed as esoteric financial instruments. In 1962, James Boness, in his PhD thesis, came up with a formula for pricing option contracts, which later laid the groundwork for the famous Black-Scholes (BS) option pricing formula.

The seminal work of Black and Scholes (1973) [2] laid the foundation for modern option pricing theory, enabling a theoretical framework for option trading and hedging. Implied volatility is one of the key inputs of this formula, which, unlike other inputs, is not directly observable in the market and has to be back-calculated using market data. One of the main advantages of the Black-Scholes model is that it provides a closed-form solution for implied volatility, which can be calculated by inverting the formula.

The fundamental assumption of the model is that the underlying asset price S_t of an option contract follows a stochastic process described by a geometric brownian motion (GBM), given by:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{2.1}$$

where μ is the drift component (the expected return of the asset), σ is the constant volatility, and W_t is a Wiener process representing Brownian motion. Using Ito's Lemma, the Black-Scholes equation for a European call option can be written as:

$$\frac{\partial C}{\partial t} + rS\frac{\partial C}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} = rC \tag{2.2}$$

where C is the price of the option contract, S is the underlying asset price, T is time to maturity, and r is the risk-free interest rate. The closed-form solution to this equation for a European call option is given by:

$$C(S,t) = SN(d_1) - Ke^{-rT}N(d_2) \tag{2.3}$$

where K is the strike price, $N(\cdot)$ is the cumulative distribution function of the standard normal distribution, and:

$$d_1 = \frac{\ln(S/K) + (r + \frac{\sigma^2}{2})(T)}{\sigma\sqrt{T}} \tag{2.4}$$

$$d_2 = d_1 - \sigma\sqrt{T} \tag{2.5}$$

In the Black-Scholes model, the volatility σ of the underlying asset is assumed to remain constant over the life of the option. However, empirical evidence has shown that this assumption does not hold in real-world financial markets. Numerous studies have demonstrated that implied volatility exhibits a phenomenon known as the "volatility smile" or "volatility skew," where implied volatility varies with the strike price and time to maturity of the option.

Challenging Constant Volatility

One of the earliest papers to challenge the constant volatility assumption was by Rubinstein (1985) [3], who observed that implied volatilities of european style S&P 500 index options exhibited a systematic pattern across strike prices. This finding was further supported by the work of Derman and Kani (1994) [4] and Dupire (1994) [5], who introduced the local volatility models, constructing a volatility surface to account for the volatility smile in options markets enhances pricing accuracy compared to the constant volatility assumption of the Black-Scholes model. These models allowed for a more flexible representation of the implied volatility surface, albeit at the cost of increased complexity.

Subsequent research by Dumas, Fleming, and Whaley (1998) [6] and Peña, Rubio, and Serna (1999) [7] provided further evidence of the non-constant nature of implied volatility and Black-Scholes assumption of log-normal price distribution of underlying asset. These studies employed regression-based techniques to model the implied volatility surface and demonstrated the presence of significant patterns and dynamics in implied volatility across different markets and time periods.

Volatility Forecasting

Volatility forecasting has emerged as a crucial area of research in finance, as accurate predictions of future volatility are essential for risk management, option pricing, and investment decision-making. Volatility forecasting refers to predicting the future volatility of a financial asset, given the previous volatility values. Over the years, various approaches have been proposed to forecast both historical and implied volatility. While historical volatility forecasting relies on past price data, implied volatility forecasting focuses on extracting market expectations from option prices. Given the forward-looking nature of implied volatility, it has gained significant attention from both academics and practitioners. Numerous studies have explored the realm of historical volatility forecasting, employing a wide range of econometric and machine learning techniques. Classical time series models such as ARCH and GARCH try to capture the autoregressive nature of volatility series.

In their 2003 paper, Poon and Granger [8] provide a comprehensive review of volatility forecasting, focusing on empirical findings from 93 papers published over two decades. The review highlights the importance of volatility in areas such as investment, option pricing, and market regulation. The authors compare the performance of different forecasting methods, particularly historical volatility models and implied volatility from options, across various asset classes and geographic regions. They also discuss challenges in volatility measurement, including data frequency and handling extreme values. Hansen and Lunde (2005) [9] conducted a benchmark study comparing over 339 volatility models against the standard ARCH(1) and GARCH(1,1) models for one-day-ahead volatility forecasting. Their findings indicate that none of the tested models outperform the GARCH(1,1) model, while the ARCH(1) model proved to be less effective than the others. The 2006 paper by Nikolay Gospodinov, Athanasia Gavala, and Deming Jiang [10] investigates the time-series properties of S&P 100 volatility and evaluates different volatility forecasting models. They examine nonparametric and parametric measures such as implied and realized volatility, finding that volatility exhibits slow mean-reverting behavior and potential long memory. By using bootstrap methods and intercept corrections, they improve forecast accuracy, particularly for the next-period implied volatility. The study highlights the importance of accounting for model misspecifications in volatility forecasts. More recently, Sévi [11] investigates the use of high-frequency intraday data to predict crude oil futures volatility over short to medium-term horizons (1 to 66 days). The study examines the decomposition of realized variance into positive/negative semivariances and continuous/discontinuous parts (jumps). Although these decompositions offer insights in in-sample predictive regressions, the paper finds that out-of-sample forecasts show limited improvement using these components. In practice, simpler autoregressive models based on past realized variances performed comparably to more sophisticated approaches. These studies have consistently demonstrated the superiority of machine learning approaches over traditional econometric models in capturing the complex and nonlinear dynamics of historical volatility. Ruoxuan Xiong, Eric P. Nichols, and Yuan Shen [12] explores the application of Long Short-Term Memory (LSTM) neural networks to forecast S&P 500 volatility. The model incorporates Google Domestic Trends data as proxies for public sentiment and macroeconomic indicators, in addition to traditional financial variables. The authors demonstrate that integrating these trends significantly improves volatility prediction, especially over short horizons. This study shows the potential of using non-traditional data sources in financial forecasting. Chuong Luong and Nikolai Dokuchaev [13] explores the application of the Random Forests algorithm to forecast the realised volatility of financial time series. It extends the traditional heterogeneous autoregressive (HAR) model by incorporating purified implied volatility and the predicted direction of future volatility. The authors demonstrate that using Random Forests leads to improved accuracy in predicting both the direction and magnitude of volatility, based on historical high-frequency data.

Early works on implied volatility forecasting primarily relied on parametric models. For instance, Christensen and NR Prabhala [14] examines the correlation between implied volatility (IV), derived from option prices, and realized volatility (RV), observed in the underlying asset. It finds that IV generally predicts future RV well, though the relationship is influenced by factors such as market conditions, liquidity, and time to maturity. Similarly, Christensen and Hansen (2002) [15] uses GARCH models, regression analysis, and Vector Autoregressive Models (VAR) to explore how implied volatility (IV) forecasts realized volatility (RV). The study confirms that IV is a significant predictor of RV but highlights that its accuracy varies with market conditions and time. The paper also notes the practical implications of understanding the deviations between IV and RV for trading strategies and risk management.

As the field progressed, researchers began to explore more sophisticated techniques to capture the complex dynamics of implied volatility. Gonçalves and Guidolin (2006) [16] applied a two-factor model to the implied volatility surface of S&P 500 index options and found evidence of predictable patterns in the dynamics of implied volatility. Konstantinidi, Skiadopoulos, and Tzagkaraki (2008) [17] explores whether the future evolution of implied volatility can be predicted using data from European and US volatility indices. They employed a principal component analysis (PCA) approach to forecast implied volatility indices, demonstrating the superiority of their method over traditional time series models.

The advent of machine learning techniques has revolutionized the field of volatility forecasting. These data-driven approaches have the ability to capture complex nonlinear relationships and have shown promising results in predicting both historical and implied volatility.

In the context of implied volatility forecasting, machine learning techniques have been applied to predict implied volatility surfaces and option prices. Audrino and Colangelo (2010) [18] introduces a forecasting method for the implied volatility surface using regression trees. Their semi-parametric approach captures non-linear dynamics and improves accuracy over traditional models. J. Hosker, S. Djurdjevic, and H. Nguyen (2018) [19] explores the application of machine learning techniques to enhance the forecasting of VIX futures. The authors utilize various machine learning models to analyze historical data and predict future VIX futures prices. Their findings demonstrate that machine learning methods significantly improve forecasting accuracy compared to traditional approaches. Recent studies have focused on leveraging the power of ensemble methods and hybrid models to enhance the accuracy of implied volatility forecasting. Liu, Oosterlee, and Bohte (2019) [20] investigates the use of neural networks for option pricing and implied volatility computation. The authors develop neural network models to estimate option prices and derive implied volatilities from these estimates. Their findings reveal that neural networks can effectively model complex pricing structures and compute implied volatilities with high accuracy, outperforming traditional methods in terms of precision and computational efficiency. Similarly, R. Culkin and S.R. Das [21] investigates using deep learning models to price options. They employ deep neural networks to handle the complexity of option pricing, using historical options data (including asset prices, strike prices, and maturities) for training. The study finds that deep learning models significantly improve pricing accuracy. Spyridon D. Vrontos and John Galakis [22] focuses on forecasting the direction of changes in implied volatility using machine learning techniques. The authors apply various machine learning models to predict whether implied volatility will rise or fall, based on historical market data and option features. Their findings show that machine learning methods, including classification algorithms, effectively capture patterns and improve the accuracy of directional forecasts compared to traditional approaches. Dash (2019) [23] investigates the implied volatility surfaces of Nifty index options on India's NSE using parametric models. The study finds a "volatility smile" for both call and put options, with minimum implied volatility at 2.3-9.0% in-the-money for calls and 10.7-29.3 % in-the-money for puts. The implied volatility surface has a minimum point for calls and a saddle point for puts, with a significant jump in implied volatility when transitioning from out-of-the-money to in-the-money. Most recently, Chen, Li, and Yu (2024) [24] proposed a two-step procedure to model the dynamic implied volatility surface (IVS) using bivariate tensor-product B-splines (BTPB) with time-dependent coefficients predicted by tree-based machine learning methods. In the first step, they construct a BTPB basis to approximate the cross-sectional structure of the IVS, representing the surface by a vector of coefficients. In the second step, they allow for time-dependent coefficients and model the dynamic coefficients using a tree-based method for added flexibility. Through simulation studies and empirical analysis on S&P 500 index options, the authors demonstrate that their approach outperforms traditional linear parametric models and nonparametric tree-based machine learning methods in capturing the dynamic structure and predicting future implied volatility surfaces.

Despite the significant advancements in implied volatility forecasting, challenges remain. The high-dimensional and dynamic nature of the implied volatility surface poses difficulties in capturing its intricate patterns. Moreover, the presence of market anomalies, such as volatility smiles and smirks, adds complexity to the forecasting task. Future research directions include the incorporation of alternative data sources, such as sentiment analysis and news sentiment, to enhance the predictive power of implied volatility models. Additionally, the development of interpretable and explainable machine learning models is crucial for gaining insights into the drivers of implied volatility and building trust among market participants. In conclusion, the field of implied volatility forecasting has witnessed significant progress, transitioning from simple parametric models to sophisticated machine learning techniques. The ability to accurately predict implied volatility has important implications for option pricing, risk management, and trading strategies. As the volume and complexity of financial data continue to grow, the integration of advanced machine learning algorithms with traditional financial models holds great promise for the future of implied volatility forecasting.

3 Data

This study focuses on the NIFTY50 index, the benchmark Indian stock market index. The NIFTY50 is a free-float market-capitalization weighted index comprising the 50 largest and most liquid Indian companies listed on the National Stock Exchange (NSE). It provides a comprehensive representation of the Indian equity market, covering 13 sectors of the economy, and is widely used for benchmarking fund portfolios, index-based derivatives, and index funds.

3.1 Data Sources and Description

The data for this study was sourced from two primary channels provided by Feedsense AI: spot data and option chain data. The spot data consists of minute-level price data for the NIFTY50 index from January 2, 2017, to May 27, 2024. It includes fields such as date, time, open, high, low, and close prices for each minute-level observation. This granular data allows for a detailed analysis of the underlying index movements.

Complementing the spot data, I also utilized minute-level option price data for both call and put options on the NIFTY50 index, spanning from February 11, 2019, to May 27, 2024. The option chain data expands on the spot data fields by including additional information crucial for options analysis: expiry date and strike price. In total, our dataset encompasses 278 unique weekly option contracts over the entire period, providing a rich landscape for studying implied volatility dynamics.

3.2 Implied Volatility Calculation

A critical step in our analysis is the calculation of implied volatilities from the option chain data. I employ the inverse solution of the Black-Scholes model, adhering to put-call parity. The Black-Scholes formula for European options is given by:

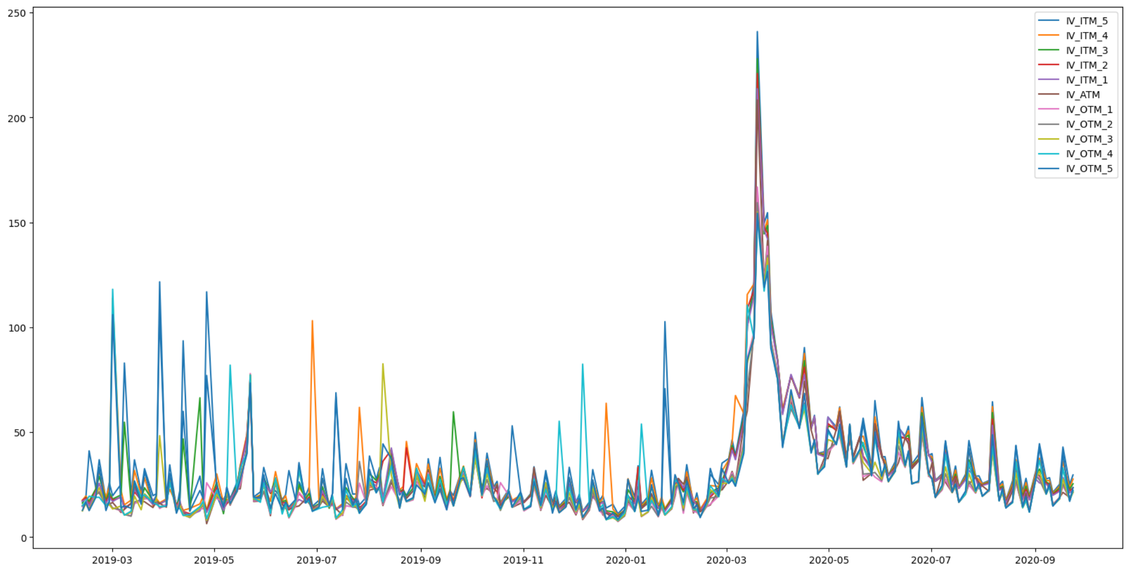


Figure 2: Time series of implied volatilities for NIFTY index options from February 2019 to October 2020. The figure shows separate series for in-the-money (ITM), at-the-money (ATM), and out-of-the-money (OTM) option contracts. The vertical axis represents implied volatility, while the horizontal axis shows the time period.

$$C = S_0 N(d_1) - K e^{-rT} N(d_2) \tag{3.1}$$

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1) \tag{3.2}$$

Where C and P represent call and put option prices respectively, S_0 is the current price of the NIFTY50 index, K is the strike price, r is the risk-free interest rate, T is time to expiration in years, σ is implied volatility, and $N(.)$ is the cumulative standard normal distribution function. The parameters d_1 and d_2 are calculated as:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T} \tag{3.3}$$

To minimize the impact of bid-ask spreads and ensure we're using the more liquid option, I employ the put option formula when the strike price is less than the current price, and the call option formula otherwise. The implied volatilities (IVs) are calculated using the Newton-Raphson method, an iterative numerical approach for approximating the roots of a real-valued function. Figure 2 shows a snapshot of different ITM, OTM and ATM time series spanning over February 2019 to October 2020.

3.3 Data Pre-processing

Our data preprocessing pipeline involved several crucial steps to ensure data quality and prepare for modeling. First, I aligned the timestamps across all datasets to ensure consistency. Then I addressed the issue of missing implied volatility values, which can occur due to no-arbitrage violations. Figure 3 shows the distribution of missing data over various strikes. Deep in-the-money and deep out-of-the-money option contracts have much more missing data compared to strikes closer to at-the-money contracts because these options are less frequently traded due to their limited intrinsic value, reduced leverage, lower liquidity, and investor preference for options with strikes closer to the current price of the underlying asset. The lower trading volume of deep ITM and deep OTM options results in reduced liquidity, wider bid-ask spreads, and higher transaction costs, making them less attractive to investors and leading to more missing data points in their time series, which can pose challenges for analyzing and modeling the implied volatility surface, particularly in regions farther away from the at-the-money strike prices. Linear interpolation technique was used to interpolate the missing IV values.

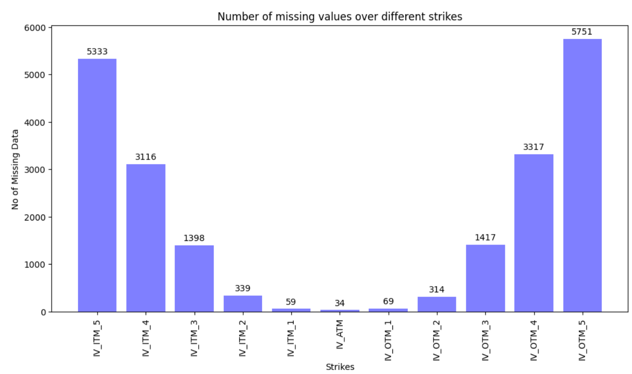


Figure 3: Distribution of missing data over different strikes

Feature engineering was done to capture important market dynamics. I calculated moneyness as $\ln(S/K)$ for each option, providing a standardized measure of how far an option is in or out of the money. Time to expiration was calculated in years, allowing for a consistent measure across different option contracts. Min-max scaling was performed to keep the input values between 0 to 1, which helps in stabilising the training of neural networks. The implied volatility smile vector is constructed using 11 dimensions, consisting of five in-the-money (ITM) strikes, one at-the-money (ATM) strike, and five out-of-the-money (OTM) strikes.

3.4 Final Dataset Structure and Summary Statistics

After preprocessing, the final dataset consists of a panel of implied volatilities across different strike prices and expiration dates for each trading day. This can be represented functionally as:

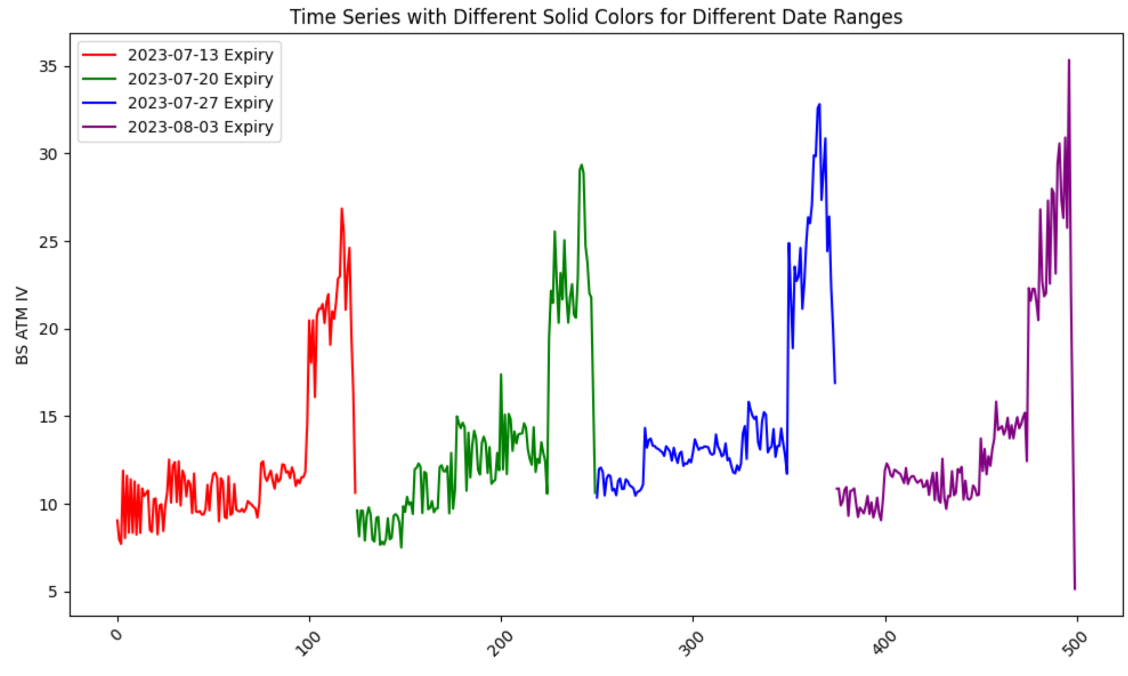


Figure 4: At-the-money IV time series for 4 successive weekly option contracts expiring on 2023-07-13, 2023-07-20, 2023-07-27 and 2023-08-03

$$IV_{t,K,T} = f(S_t, K, T, r_t, ...) \tag{3.4}$$

Where $IV_{t,K,T}$ is the implied volatility at time t for strike price K and time-to-expiration T, S_t is the NIFTY50 index level at time t, and r_t is the risk-free rate at time t. Figure 4 depicts the IV time series for 4 successive ATM contracts expiring on 13th July 2023, 20th July 2023, 27th July 2023 and 3rd August 2023.

Table 1 provides summary statistics for the implied volatilities in our dataset:

Figure 1b illustrates a sample implied volatility surface from our dataset formed by a collection of volatility smiles over a certain time period.

This surface provides a visual representation of how implied volatility varies across different strike prices and time to expiration, capturing both the volatility smile and term structure effects. Figure 5 gives a detailed visual representation of the multi-dimensional structure of the IVS data. The IV surfaces for successive independent contracts are stitched together to create a continuous IV time series.

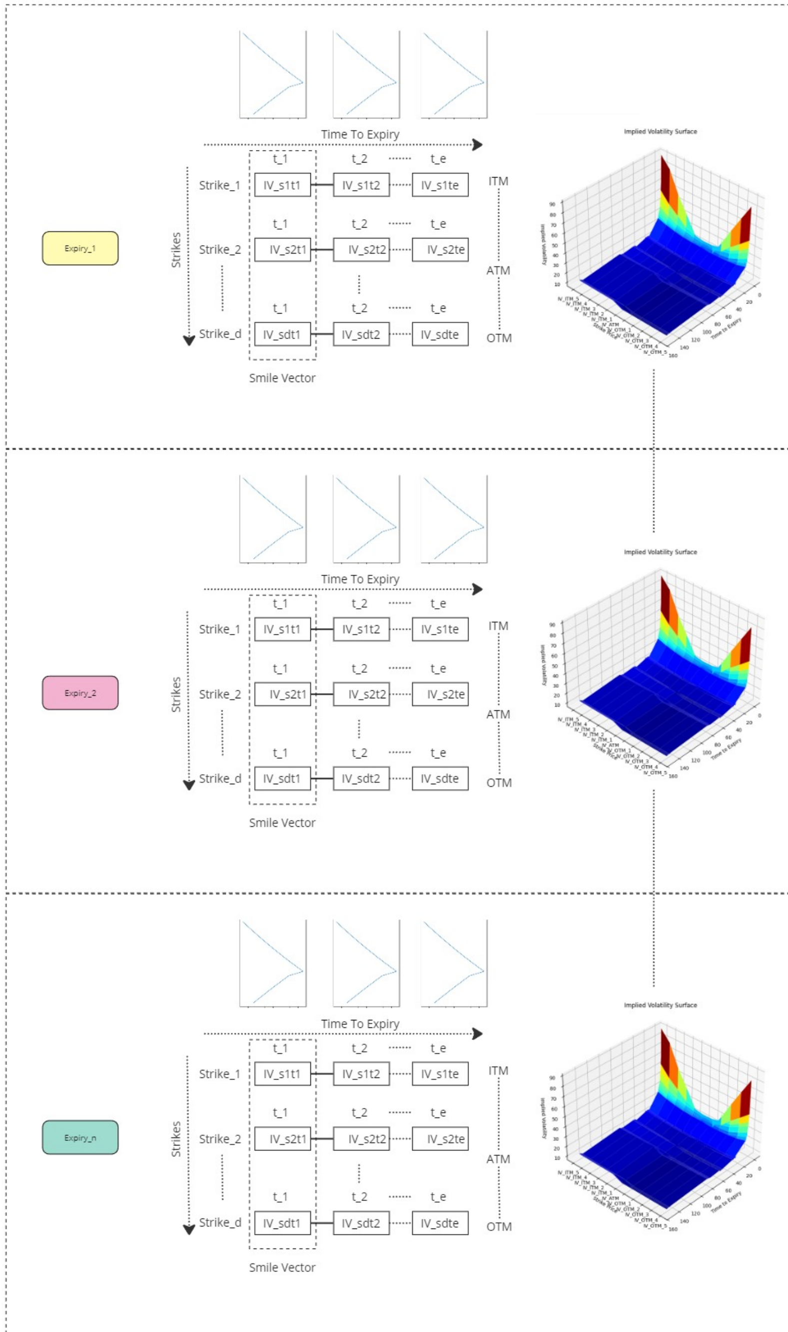


Figure 5: Structure of the implied volatility dataset for NIFTY index options. Multiple implied volatility surfaces (IVS) for different expiration dates are stitched together 'head-to-toe'. Each IVS is represented as a 2D matrix with d rows corresponding to d different strike prices (spanning out-of-the-money, at-the-money, and in-the-money options) and e columns corresponding to e different times-to-expiry. The complete dataset forms a 3D structure across strikes, times-to-expiry, and expiration dates.

Strikes	count	mean	std	min	25%	50%	75%	max
IV ITM 5	32286.00	28.83	21.01	8.77	16.80	22.30	32.86	288.80
IV ITM 4	32286.00	26.96	19.13	8.38	16.04	21.30	30.87	291.09
IV ITM 3	32286.00	25.23	17.32	8.08	15.45	20.46	29.05	292.75
IV ITM 2	32286.00	23.79	16.17	7.80	14.96	19.71	27.33	300.94
IV ITM 1	32286.00	22.69	15.50	7.27	14.56	18.99	25.85	305.69
IV ATM	32286.00	21.11	14.50	2.81	13.66	17.74	24.16	325.17
IV OTM 1	32286.00	20.33	13.13	5.45	13.41	17.27	23.57	210.90
IV OTM 2	32286.00	20.89	13.71	5.75	13.40	17.33	24.17	208.41
IV OTM 3	32286.00	21.80	14.90	5.88	13.44	17.48	25.02	207.56
IV OTM 4	32286.00	23.08	16.85	5.97	13.35	17.75	26.40	206.80
IV OTM 5	32286.00	24.50	18.69	6.18	13.90	18.24	28.13	229.05

Table 1: Descriptive statistics of the implied volatility data for the NIFTY50 index options.

3.5 Stationarity check

Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller (ADF) test is a crucial statistical tool in time series analysis, used to determine whether a unit root is present in a time series sample. This test is particularly important in financial modeling as it helps identify whether a series is stationary or not, which is fundamental for many forecasting and modeling techniques. A stationary series has constant statistical properties over time, making it more predictable and suitable for various statistical models.

I conducted ADF tests on our NIFTY50 index price and implied volatility series across different moneyness levels. Table 2 presents the ADF test statistics and corresponding p-values.

The ADF test for the NIFTY50 index price series yields a p-value of 7.195×10^{-1}, greater than the 0.05 significance level, suggesting non-stationarity and alignment with the efficient market hypothesis. In contrast, all implied volatility series across different moneyness levels show strong evidence of stationarity, with p-values significantly below 0.05. This indicates mean reversion, a fundamental property of volatility data. Mean reversion implies that periods of high volatility are likely followed by lower volatility, and vice versa, providing a basis for forecasting future volatility levels, which is crucial for option pricing models and trading strategies.

3.6 Data Limitations

While our dataset is comprehensive, it's important to acknowledge certain limitations. A key limitation of this study is the focus on short-term option contracts with a maximum of 7 days

Variable	ADF Statistic	p-value
Price	-1.089	7.195e-01
IV_ITM_5	-8.530	1.045e-13
IV_ITM_4	-7.638	1.935e-11
IV_ITM_3	-6.539	9.426e-09
IV_ITM_2	-4.158	7.760e-04
IV_ITM_1	-3.399	1.101e-02
IV_ATM	-3.806	2.848e-03
IV_OTM_1	-3.800	2.908e-03
IV_OTM_2	-4.275	4.906e-04
IV_OTM_3	-6.198	5.900e-08
IV_OTM_4	-8.423	1.965e-13
IV_OTM_5	-8.703	3.755e-14

Table 2: Augmented Dickey-Fuller Test Results

to expiration. While these contracts are highly liquid during this period, I may be missing valuable historical information about their dynamics. Option contracts typically become liquid around 3 months before expiration, but by limiting our analysis to the final week, I potentially overlook important patterns and trends that develop over longer timeframes. This constraint may impact the model's ability to capture longer-term factors influencing implied volatility surfaces. Deep out-of-the-money options may have less reliable implied volatilities due to lower liquidity, potentially introducing a bias in these regions of the volatility surface. The use of interpolation for missing data points, while necessary, may smooth out some features of the volatility surface. Lastly, while care was taken to ensure that only information available at each point in time was used in the modeling process, subtle forms of look-ahead bias may still be present.

4 Method

The method section presents the mathematical formulation and computational approaches employed in this study to model and predict implied volatility surfaces (IVS) using traditional machine learning techniques. The primary focus is on comparing the performance of traditional econometric models, such as Vector AutoRegression (VAR) and Vector Error Correction Model (VECM), with advanced deep learning architectures, specifically Long Short-Term Memory (LSTM) and Attention-based LSTM (AttLSTM) models.

4.1 Problem Formulation

Let $S = \{s_1, s_2, \ldots, s_t\}$ be a set of implied volatility (IV) smiles observed over the last t timestamps, where each S_i represents the IV smile at timestamp i. The goal is to predict the IV smile at the next timestamp, s_{t+1}, given the historical data S.

Each IV smile S_i is an 11-dimensional vector containing the implied volatilities for 11 different strike prices:

$$s_i = [IV_{1i}, IV_{2i}, \ldots, IV_{11i}]$$

The implied volatility for the j^{th} strike price at timestamp i is denoted as IV_{ji}, where $j \in 1, 2, \ldots, 11$ represents the index of the strike price, and $i \in 1, 2, \ldots, t$ represents the timestamp. In this case, $t = 30$, which means I am considering the last $30 \times 15, \text{min} = 450, \text{mins}$ of IV data to forecast the next set of IV smiles. The 11 strike prices are categorized as follows: 5 In-The-Money (ITM) strikes: $IV_{1i}, IV_{2i}, IV_{3i}, IV_{4i}, IV_{5i}$; 1 At-The-Money (ATM) strike: IV_{6i}; and 5 Out-of-The-Money (OTM) strikes: $IV_{7i}, IV_{8i}, IV_{9i}, IV_{10i}, IV_{11i}$.

The historical IV smile data can be represented as a matrix $\tilde{S}$:

$$\tilde{S} = \begin{bmatrix} S_1^\top \\ S_2^\top \\ \vdots \\ S_t^\top \end{bmatrix}$$

where each row of $\tilde{S}$ corresponds to an IV smile at a specific timestamp, and each column represents the implied volatilities for a particular strike price across all timestamps.

To formulate the prediction problem mathematically, I define a function f that maps the historical IV smiles to the next timestamp's or set of timestamp's IV smile(s):

$$s_{t+1} = f(s_1, s_2, \ldots, s_t)$$

This is a multi-variate single/multi-step refression problem and the regression function $f(\cdot)$ can be modeled using various techniques mentioned above.

The following subsections provide detailed descriptions of each model, their underlying assumptions and training procedures.

4.2 Baseline Models

4.2.1 Vector Auto Regression (VAR) Model

Vector AutoRegression (VAR) is a multivariate time series model that captures the linear interdependencies among multiple variables over time. In the context of implied volatility surface modeling, VAR assumes that the future values of implied volatilities across different strike prices and maturities can be predicted based on their own lagged values and the lagged values of other variables in the system.

The VAR model of order $k = 30$, denoted as VAR(30), for our IV forecasting problem can be expressed as:

$$\hat{s}_{t+1} = c_t + \Phi_1 s_t + \Phi_2 s_{t-1} + \cdots + \Phi_k s_{t-k} + \epsilon_t \tag{4.1}$$

where:

- $\hat{s}_{t+1} \in \mathbb{R}^{1\times 11}$ is the forecasted implied volatility smile at time t and s_{t-i} is the i^{th} time lag value of the same,
- $c_t = (c_1, c_2, \ldots, c_{11})$ is a vector of constants,
- Φ_j, is the j^{th} coefficient of the lagged variables,
- $\epsilon_t = (\epsilon_1, \epsilon_2, \ldots, \epsilon_{11})$ is a vector of white noise error terms.

The VAR model is estimated using ordinary least squares (OLS) on each equation of it's respective variable separately. The optimal lag order is identified using information criteria like the Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC). After estimating the model, it can be utilized to forecast future values of the implied volatility variables.

The Vector Autoregression (VAR) model serves as a solid baseline for modeling the dynamics of the implied volatility surface, capturing the linear interdependencies among the implied volatilities for different log-moneyness groups and expiration months. However, the VAR model has limitations when dealing with non-stationary time series that exhibit long-term equilibrium relationships, known as cointegration. In the context of the implied volatility surface, significant cointegrating relationships have been observed between the log-moneyness groups and expiration months, suggesting the presence of a common stochastic trend. While the VAR model captures the short-term dynamics, it does not explicitly account for these long-term relationships, potentially leading to suboptimal forecasts. To address this limitation, the Vector Error Correction Model (VECM) emerges as a potential improvement over the VAR model. The VECM extends the VAR model by incorporating an error correction term that explicitly accounts for the cointegrating relationships, allowing for a more accurate representation of the underlying dynamics and improved forecast accuracy, especially over longer horizons.

4.2.2 Vector Error Correlation Model (VECM)

The Vector Error Correction Model (VECM) is an extension of the VAR model that incorporates cointegration relationships among the variables. Cointegration occurs when two or more time series have a long-run equilibrium relationship. The VECM captures both the short-run dynamics and the long-run equilibrium adjustments in the system.

The VECM can be expressed as:

$$\Delta s_t = c + \Pi s_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta s_{t-j} + \epsilon_t$$

where:

- Δs_t is the first difference of the implied volatility vector s_t,
- $\Pi = \alpha\beta'$ is the long-run impact matrix, with α representing the speed of adjustment to equilibrium and β representing the cointegrating vectors,
- Γ_j are short-run coefficient matrices,
- ϵ_t is a vector of white noise error terms.

The VECM is estimated using the Johansen procedure, which involves the following steps:

1. Test for the presence and rank of cointegration using the trace test or the maximum eigenvalue test.
2. Estimate the cointegrating vectors β using maximum likelihood estimation.
3. Estimate the speed of adjustment coefficients α and the short-run dynamics Γ_j using OLS.

The interpretation of the VECM parameters provides insights into the long-run equilibrium relationships and the short-run adjustments in the implied volatility surface dynamics.

4.3 Long short Term Memory (LSTM) network

Long Short-Term Memory (LSTM) neural networks, introduced by Hochreiter and Schmidhuber (1997) [25], are a variant of Recurrent Neural Networks (RNNs) that have gained prominence in sequence learning tasks. RNNs are designed to handle sequential data by maintaining a hidden state that carries information across time steps, allowing the model to capture temporal dependencies. However, traditional RNNs suffer from the vanishing and exploding gradient problems, which hinder their ability to learn long-term dependencies.

LSTMs overcome these limitations by using gating mechanisms to control the flow of information within the network. The key components of an LSTM cell are the input gate, forget gate, and output gate, which control the updating, retention, and output of the cell state. These gates enable LSTMs to selectively remember or forget information over long sequences, making them well-suited for tasks involving time series data, such as language modeling, speech recognition, and financial forecasting [26, 27, 28].

The mathematical formulation of an LSTM cell can be described as follows:

$$\text{Input gate:} \quad i_t = \sigma(W_i[h_{t-1}, s_t] + b_i) \tag{4.2}$$

$$\text{Forget gate:} \quad f_t = \sigma(W_f[h_{t-1}, s_t] + b_f) \tag{4.3}$$

$$\text{Output gate:} \quad o_t = \sigma(W_o[h_{t-1}, s_t] + b_o) \tag{4.4}$$

$$\text{Candidate memory state:} \quad \tilde{c}_t = \tanh(W_c[h_{t-1}, s_t] + b_c) \tag{4.5}$$

$$\text{Memory state update:} \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{4.6}$$

$$\text{Hidden state update:} \quad h_t = o_t \odot \tanh(c_t) \tag{4.7}$$

where:

- s_t is the input vector at time step t,
- h_t and c_t are the hidden state and cell state vectors at time step t, respectively,
- W and b are the weight matrices and bias vectors for the respective gates and candidate memory state,
- σ and tanh are the sigmoid and hyperbolic tangent activation functions, and
- $\odot$ denotes element-wise multiplication.

The LSTM model is implemented using the PyTorch framework. The model consists of an LSTM layer followed by a dropout layer and a linear transformation. The LSTM layer takes an input sequence of shape $(\text{batch\_size}, n_{\text{lookback}}, n_{\text{features}})$, where n_{lookback} is the number of historical time steps considered, and n_{features} is the number of input features (11 in this case). The hidden state dimension is set to 512.

The forward pass of the model involves the following steps:

1. The input sequence is passed through the LSTM layer, which outputs the hidden states for each time step.
2. The dropout layer is applied to the LSTM outputs to regularize the model and prevent overfitting.
3. The last hidden state of the LSTM is selected and passed through a linear transformation to obtain the output of shape $(\text{batch\_size}, n_{\text{forecast}} \times n_{\text{features}})$, where n_{forecast} is the number of future time steps to predict.
4. The output is reshaped to $(\text{batch\_size}, n_{\text{forecast}}, n_{\text{features}})$ to represent the predicted sequence.

While the presented LSTM model provides a powerful framework for capturing temporal dependencies, it has some limitations. One potential issue is the fixed-length input sequence, which may not be optimal for all time series patterns. Additionally, the model does not explicitly consider the spatial dependencies among the input features, which could be crucial in the context of implied volatility surface modeling.

To address these limitations, I've built an attention-based LSTM network that can dynamically focus on relevant parts of the input sequence, regardless of their position. This attention mechanism allows the model to weigh the importance of different time steps and features, potentially capturing both short-term fluctuations and long-term trends in the implied volatility surface. Additionally, the attention layer can implicitly model spatial

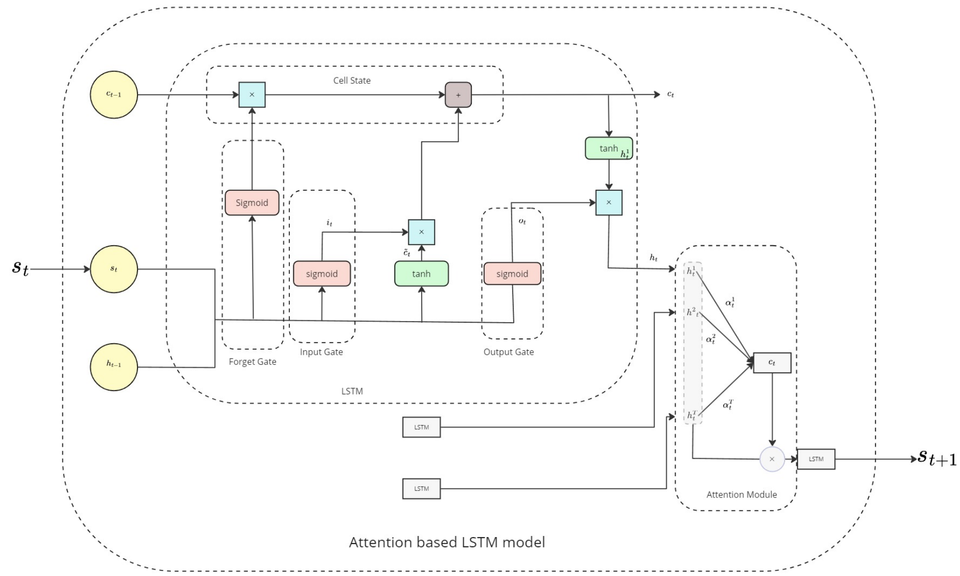


Figure 6: Overall architecture of the attention-based long short-term memory (LSTM) model.

relationships between different strikes and maturities, helping to better capture the complex interactions across the entire volatility surface.

4.4 Attention based Long short Term Memory (Att-LSTM) network

Attention mechanisms, introduced by Vaswani et al. (2017) [29] have gained significant popularity in various tasks, such as machine translation, image captioning, and speech recognition [30, 31, 32] Inspired by the human visual system, attention allows a model to selectively focus on important parts of the input with varying degrees of emphasis, guided by attention weights. This mechanism helps to address the fixed context length issue of traditional neural networks and enables the model to learn complex patterns in the data.

As shown in figure 6 the Attention-based LSTM (AttLSTM) model extends the standard LSTM architecture by incorporating attention mechanisms to focus on relevant parts of the input sequence when making predictions [30, 33]. In the context of implied volatility surface modeling, the attention mechanism allows the model to assign different weights to different strike prices and maturities based on their importance in predicting future implied volatilities.

The AttLSTM model consists of two main components: an LSTM encoder and an attention layer. The LSTM encoder processes the input sequence and generates a sequence of hidden states h_t. The attention layer then computes a weighted sum of the hidden states, where the weights α_t are determined by the compatibility between each hidden state h_t and a learnable query vector v.

The key equations for the AttLSTM model are:

$$\begin{aligned}
\text{LSTM encoder:} \quad & h_t = \text{LSTM}(s_t, h_{t-1}) \\
\text{Attention weights:} \quad & \alpha_t = \text{softmax}\left(v^T(W_a[h_t; c])\right) \\
\text{Context vector:} \quad & c_t = \sum_t \alpha_t h_t \\
\text{Attention output:} \quad & a_t = \tanh(W_c[c_t; h_t])
\end{aligned}$$

where:

- h_t is the hidden state vector at time step t,
- s_t is the input vector at time step t,
- v, W_a, and W_c are learnable parameters,
- $[;]$ denotes concatenation,
- softmax is the softmax activation function, defined as:

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

The AttLSTM model is trained using the backpropagation through time (BPTT) algorithm [34] and stochastic gradient descent optimization. The attention weights are learned jointly with the LSTM parameters during the training process.

The architecture consists of an LSTM encoder followed by an attention layer, a dropout layer, and a fully connected output layer. The attention mechanism is implemented using the multiplicative attention scheme, where the compatibility between the hidden states and the query vector is computed using a feedforward layer:

$$\text{score}(h_t, v) = v^T(W_a h_t)$$

The AttLSTM model has several advantages over the standard LSTM model. By incorporating attention, the model can adaptively focus on the most relevant parts of the input sequence, leading to improved prediction accuracy. Additionally, the attention weights provide interpretability by indicating which strike prices and maturities are most important for predicting future implied volatilities.

In summary, the AttLSTM model combines the powerful sequence modeling capabilities of LSTMs with the selective focus of attention mechanisms. This allows the model to effectively capture the complex dynamics of the implied volatility surface while emphasizing the most relevant information for making predictions.

4.5 Training and Optimization

In this research, I adhere to best practices for training and optimizing machine learning models. To maintain the temporal order crucial for time series data, the dataset is divided into train, test, and validation splits without shuffling. Although minute-level option chain data is available, due to computational constraints, the data is first sampled into 15-minute intervals. Most of the experiments are conducted using this 15-minute interval data.

The entire dataset is divided into train, test, and validation splits in a 70%-20%-10% ratio. Data from July 2, 2019, to June 6, 2022, is used to train the models, while data from June 6, 2022, onwards is used for validation and testing. To ensure a fair comparison between the baseline and deep learning-based models, the same data split is used during the training of all models. I use Python's `statsmodel`[2] library to build the baseline models and `PyTorch`[3], a popular deep learning framework, to build, train, and optimize the deep learning models.

The data is transformed to suit multivariate supervised time series learning algorithms by dividing it into an input group X: $X_{t-n_{\text{lookback}}}, X_{t-n_{\text{lookback}}+1}, \ldots, X_{t-1}$, and an output group Y: $X_t, X_{t+1}, \ldots, X_{t+n_{\text{forecast}}-1}$. The variable n_{lookback} indicates the number of previous time steps the model considers before making a prediction, while n_{forecast} represents the number of time steps ahead the model can predict. For $n_{\text{forecast}} = 1$, it is a multivariate single-step time series forecasting problem, and for $n_{\text{forecast}} > 1$, it is a multivariate multi-step time series forecasting problem. For most experiments, n_{lookback} is set to 30 unless otherwise mentioned. The use of an n_{lookback} window allows the model to capture temporal dependencies and patterns in the data, which can improve prediction accuracy.

All IV values in the dataset are scaled between 0 and 1 to normalize the data and prevent certain features from dominating the learning process. Scaling the data also helps improve the convergence speed and stability of the optimization algorithms used to train the models.

The weights of the LSTM models are initialized using the He-normal initialization method [35] and are trained for 1,000 epochs. To reduce model overfitting, early stopping, gradient clipping, and dropouts are employed. A cosine learning rate scheduler is used with an initial learning rate of 0.001, and a batch size of 128 is used. As this is a regression problem, mean squared error (MSE) is chosen as the loss function. The Adam optimizer [36] is used for training the models.

During training, model weights are only saved if the performance of the model on the validation data exceeds the previous best performance. Additionally, the last model weights are saved to allow for resuming training in the future if necessary. This approach ensures

[2]https://www.statsmodels.org/stable/index.html
[3]https://pytorch.org/

that the best-performing model is selected based on its performance on unseen validation data, which helps to prevent overfitting and improves the model's generalization ability.

5 Results and Discussion

In this section, I present the results of various methods for the task of forecasting implied volatility on our NIFTY option chain data and compare their efficacy over different evaluation metrics. Our primary objective is to investigate how well deep learning models, such as the LSTM model, capture the implied volatility surface (IVS) of one of the most traded Indian indices and whether they are superior to traditional time series models like VAR and VECM. Additionally, I examine whether the attention mechanism can help filter out vital information, thus improving the IVS modeling performance. I also aim to address the following questions:

- How does the performance of these models change when making longer-horizon forecasts compared to shorter ones?
- How do the model architecture and complexity contribute to their performance?

To account for longer-horizon forecasts, all models are trained separately to predict single and multi-step outputs of 1, 4, 8, and 16 steps ahead of the observed 15-minute timeframe data, resulting in 15-minute, 1-hour, 2-hour, and 4-hour forecasts. To ensure fair comparisons, all models are evaluated on the same test dataset. Furthermore, the results of both in-sample (training data) and out-of-sample (test data) are reported. In-sample results are useful for understanding a model's ability to fit a particular problem, whereas out-of-sample results help to understand the generalization capability of a model to new data.

The models are evaluated using five different evaluation metrics, providing a comprehensive overview of their performance for the task of IV forecasting.

5.1 Evaluation Metrics

In this thesis, I employ several evaluation metrics to assess the performance of the proposed methods for multistep time series prediction of implied volatility surfaces. These metrics provide a comprehensive understanding of the accuracy and effectiveness of the models in capturing the dynamics of the implied volatility surface. The evaluation metrics used in this study are following:

Mean Absolute Error (MAE)

Mean Absolute Error is a widely used metric that measures the average magnitude of the absolute differences between the predicted and actual values. In the context of multistep time series prediction, MAE is calculated by taking the mean of the absolute errors across all time steps and samples. The mathematical equation for MAE is given by:

$$\text{MAE} = \frac{1}{N \cdot T} \sum_{i=1}^{N} \sum_{t=1}^{T} \left| \hat{IV}_{it} - IV_{it} \right| \tag{5.1}$$

where N is the number of samples, T is the number of time steps in the prediction horizon, $\hat{y}_{it}$ is the predicted value for sample i at time step t, and y_{it} is the corresponding actual value. MAE provides a clear interpretation of the average prediction error in the same units as the original data.

Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error is a widely used metric for evaluating the accuracy of forecasting models, particularly in the context of time series data. MAPE measures the average absolute percentage difference between the predicted and actual values, providing an intuitive interpretation of the model's performance in percentage terms.

The mathematical equation for MAPE in the context of multistep time series prediction is:

$$\text{MAPE} = \left(\frac{100\%}{N \cdot T} \right) \sum_{i=1}^{N} \sum_{t=1}^{T} \frac{|\hat{IV}_{it} - IV_{it}|}{|IV_{it}|}$$

where $\hat{IV}_{it}$ is the predicted value for the i-th time series at time t, IV_{it} is the actual value for the i-th time series at time t, N is the number of time series, and T is the number of time steps in the forecasting horizon.

MAPE is useful for quantifying error in percentage terms, making it scale-independent and suitable for comparing different models or evaluating a model's performance across various data scales. A lower MAPE indicates better accuracy, with 0% representing a perfect forecast. However, MAPE is undefined for actual values of zero and can be sensitive to small denominators, with a tendency to overemphasize positive errors.

Root Mean Squared Error (RMSE)

Root Mean Squared Error is the square root of the Mean Squared Error. It is a widely used metric that measures the average magnitude of the errors, providing an interpretation in the same units as the original data. The mathematical equation for RMSE in the context of multistep time series prediction is:

$$\text{RMSE} = \sqrt{\frac{1}{N \cdot T} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(\hat{IV}_{it} - IV_{it} \right)^2} \tag{5.2}$$

RMSE is often preferred over MSE because it is more interpretable and has the same units as the target variable.

Coefficient of Determination (R^2)

The Coefficient of Determination, denoted as R^2, is a statistical measure indicating the proportion of variance in the dependent variable that can be predicted from the independent variables. In the context of multistep time series prediction, R^2 is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{N} \sum_{t=1}^{T} (IV_{it} - \hat{IV}_{it})^2}{\sum_{i=1}^{N} \sum_{t=1}^{T} (IV_{it} - \bar{IV})^2} \tag{5.3}$$

where $\bar{IV}$ is the mean of the actual values. R^2 ranges from 0 to 1, with a higher value indicating a better fit of the model to the data. An R^2 of 1 indicates that the model perfectly explains the variance in the dependent variable, while an R^2 of 0 suggests that the model does not explain any of the variance.

Mean Directional Accuracy (MDA)

Mean Directional Accuracy is a metric that evaluates the ability of a model to predict the correct direction of change in the time series. In the context of implied volatility surface prediction, MDA measures the proportion of correctly predicted directions (up or down) of the implied volatility changes across all time steps and samples. The mathematical equation for MDA is:

$$\text{MDA} = \frac{1}{N \cdot T} \sum_{i=1}^{N} \sum_{t=1}^{T} I\left(\text{sign}(\hat{IV}_{it} - IV_{i,t-1}) = \text{sign}(IV_{it} - IV_{i,t-1})\right) \tag{5.4}$$

where $I(\cdot)$ is an indicator function that returns 1 if the condition is true and 0 if it is false and $\text{sign}(\cdot)$ is a function that returns the sign of a number (-1 for negative, 0 for zero, and 1 for positive). MDA is particularly useful in financial applications, where the direction of change in implied volatility can be more important than the magnitude of the change itself.

By considering multiple metrics, we can gain insights into different aspects of the models' predictive capabilities, such as accuracy, error magnitude, and directional correctness.

5.2 Quantitative Results

I begin by assessing the single-step forecast performance of all our models. Trained on 15-minute interval IV data, the models predict the appearance of the IV smile in the next 15 minutes. Table 3 presents the single-step forecasting performance evaluated using five key metrics for both in-sample and out-of-sample data.

For the in-sample results, the LSTM and Att-LSTM models achieve an impressive R^2 value of 0.91, indicating that they can explain 91% of the variance in the implied volatility. In contrast, the VAR and VECM models have significantly lower R^2 values. The deep learning models also outperform the baseline models in terms of RMSE, MAPE, and MAE, with the

	In-Sample Results					Out-of-sample Results				
Method	RMSE	MAPE	MAE	R2	MDA	RMSE	MAPE	MAE	R2	MDA
VAR	0.012	1.77	0.0082	0.38	0.54	0.0292	2.20	0.0095	0.22	0.52
VECM	0.011	1.33	0.0068	0.40	55	0.028	2.08	0.0094	0.29	0.52
LSTM	0.007	0.87	0.0020	0.91	0.57	0.013	0.92	0.0050	0.85	0.53
Att-LSTM	0.006	0.42	0.0016	0.91	0.61	0.0086	0.45	0.0038	0.92	0.58

Table 3: Mean out-of-samples performance of different models evaluated over 15 minute (single-step prediction) time horizon

Att-LSTM model achieving the lowest errors.

The out-of-sample results follow a similar pattern, with the Att-LSTM model demonstrating the best performance across all metrics, closely followed by the LSTM model. The VAR and VECM models show significantly weaker performance. These results clearly indicate that our deep learning models, particularly the Att-LSTM, fit the training data much better and have superior generalization capability compared to the baseline models.

Tables 4, 5, and 6 summarize the forecast performance of these models on 1-hour, 2-hour, and 4-hour forecasting windows, respectively. As expected, the performance of all models declines as the forecasting horizon increases due to the accumulation of errors and increased uncertainty associated with predicting further into the future.

	In-Sample Results					Out-of-sample Results				
Method	RMSE	MAPE	MAE	R2	MDA	RMSE	MAPE	MAE	R2	MDA
VAR	0.026	1.96	0.0096	0.26	0.54	0.0382	2.82	0.0118	0.20	0.52
VECM	0.024	1.94	0.0092	0.30	0.54	0.0375	2.60	0.0102	0.23	0.51
LSTM	0.005	0.72	0.006	0.88	0.58	0.013	0.84	0.009	0.83	0.55
Att-LSTM	0.004	0.70	0.006	0.90	0.59	0.012	0.60	0.006	0.85	0.57

Table 4: Mean out-of-samples performance of different models evaluated over 1 hour (multi-step prediction) time horizon

For the 1-hour forecasting window (Table 4), the Att-LSTM model maintains its lead, with the best out-of-sample results, closely followed by the LSTM model. The VAR and VECM models show a more significant performance drop compared to their single-step forecast results.

Similar trends are observed for the 2-hour (Table 5) forecasting windows. The Att-LSTM model consistently outperforms the other models, albeit with a slight decline in performance as the horizon increases. Despite this decline, the Att-LSTM model still maintains a clear advantage over the other models, particularly the baseline VAR and VECM models, which show a more substantial performance deterioration. For the 4-hour (Table 6) forecasting

Method	In-Sample Results					Out-of-sample Results				
	RMSE	MAPE	MAE	R2	MDA	RMSE	MAPE	MAE	R2	MDA
VAR	0.030	2.87	0.014	0.20	0.52	0.052	3.42	0.024	0.16	0.51
VECM	0.032	2.88	0.016	0.21	0.56	0.054	3.45	0.020	0.17	0.52
LSTM	0.009	0.87	0.007	0.79	0.57	0.016	1.253	0.009	0.734	0.54
Att-LSTM	0.008	0.85	0.006	0.81	0.57	0.014	0.91	0.008	0.77	0.553

Table 5: Mean out-of-samples performance of different models evaluated over 2 hour (multi-step prediction) time horizon

Method	In-Sample Results					Out-of-sample Results				
	RMSE	MAPE	MAE	R2	MDA	RMSE	MAPE	MAE	R2	MDA
VAR	0.084	3.76	0.040	0.16	0.52	0.128	4.82	0.045	0.12	0.50
VECM	0.085	3.66	0.038	0.17	0.55	0.106	4.62	0.046	0.11	0.52
LSTM	0.014	1.22	0.008	0.71	0.60	0.02	1.45	0.011	0.67	0.52
Att-LSTM	0.018	1.24	0.008	0.68	0.58	0.02	1.79	0.012	0.59	0.52

Table 6: Mean out-of-samples performance of different models evaluated over 4 hour (multi-step prediction) time horizon

window however the LSTM model took the lead in terms of MAPE, MAE and R2 values. This result can be attributed to its ability to capture long-term dependencies in the data. As the forecasting horizon increases, the attention mechanism in the Att-LSTM model may struggle to focus on the most relevant information from the distant past, while the LSTM model's simpler architecture allows it to maintain a better balance between capturing long-term dependencies and avoiding overfitting.

Comparing the in-sample and out-of-sample results across all forecasting horizons, I observe that the deep learning models (LSTM and Att-LSTM) exhibit a smaller performance gap between the two sets compared to the baseline models (VAR and VECM). This finding suggests that the deep learning models are less prone to overfitting and have better generalization capabilities.

This pattern persists for the multi-step forecasts, with the Att-LSTM model maintaining a relatively smaller performance gap between in-sample and out-of-sample results compared to the baseline models.

Across all forecasting horizons and performance metrics, the Att-LSTM model consistently outperforms the other models. The LSTM model closely follows, while the VAR and VECM models show significantly weaker performance.

The R^2 metric further highlights the superiority of the Att-LSTM model. In the single-step forecast (Table 3), the Att-LSTM model achieves an out-of-sample R^2 of 0.92, indicating that it can explain 92% of the variance in the implied volatility. This value decreases for longer forecasting horizons but still remains significantly higher than the R^2 values of the other models.

The MDA metric also favors the Att-LSTM model. For the single-step forecast (Table 3), the Att-LSTM model achieves an out-of-sample MDA of 0.58, indicating that it correctly predicts the direction of change in 58% of the cases. While this value decreases for longer forecasting horizons, the Att-LSTM model maintains its lead over the other models.

Next, I shift our focus to examine how the performance of these models varies across different strike prices. For this analysis, I choose Mean Directional Accuracy (MDA) as our metric of consideration. MDA is particularly relevant in this context as it measures the models' ability to correctly predict the direction of implied volatility movements, which is crucial for option traders and risk managers who often make decisions based on directional forecasts rather than precise point estimates.

Table 7 presents the MDA results for all models across different strike prices, from deep in-the-money (ITM) to deep out-of-the-money (OTM) options. The results reveal several interesting patterns. Firstly, all models demonstrate the highest MDA for at-the-money (ATM) options, with the Att-LSTM model showing a particularly strong performance (0.681 MDA). This suggests that predicting directional movements for ATM options is generally easier, possibly due to their higher liquidity and information content. Secondly, the Att-LSTM model consistently outperforms other models across all strikes, indicating its superior ability to capture the complex dynamics of implied volatility surfaces. The performance gap is most pronounced for ATM and near-the-money options, where the attention mechanism likely helps in focusing on the most relevant features.

Interestingly, the models' performance tends to decrease for deep ITM and OTM options, with slightly better accuracy for ITM compared to OTM options. This pattern could be attributed to the lower liquidity and higher noise in the implied volatilities of these extreme strikes.

In conclusion, the quantitative results demonstrate the superiority of deep learning models, particularly the attention-based LSTM, in forecasting implied volatility surfaces for both single-step and multi-step horizons. The attention mechanism proves to be a valuable addition, enabling the model to focus on the most relevant information and improve its generalization capability. The Att-LSTM model consistently outperforms the LSTM, VAR, and VECM models across all forecasting horizons and performance metrics, making it the most suitable choice for implied volatility forecasting tasks.

5.3 Qualitative Results

IV Time Series Prediction

Figure 7 compares the actual and predicted implied volatility (IV) time series for the at-the-money (ATM) contract using four different models: VAR, VECM, LSTM, and Att-LSTM. The

Strike	VAR	VCEM	LSTM	Att-LSTM
IV_ITM_5	0.516	0.514	0.512	0.574
IV_ITM_4	0.518	0.520	0.531	0.569
IV_ITM_3	0.516	0.517	0.504	0.575
IV_ITM_2	0.518	0.521	0.509	0.576
IV_ITM_1	0.514	0.522	0.513	0.553
IV_ATM	0.572	0.586	0.581	0.681
IV_OTM_1	0.530	0.532	0.518	0.567
IV_OTM_2	0.512	0.510	0.514	0.578
IV_OTM_3	0.513	0.523	0.529	0.552
IV_OTM_4	0.517	0.510	0.540	0.575
IV_OTM_5	0.525	0.521	0.508	0.548

Table 7: Mean Directional Accuracy (MDA) of all the models over different strikes for single step forecast

predictions are made on the out-of-sample data over four forecast horizons: 15 minutes, 1 hour, 2 hours, and 4 hours. The orange lines represents the actual IV time series, while the blue line represents the predicted series.

From the time series plots, it is evident that the deep learning models (LSTM and Att-LSTM) produce predictions that more closely follow the actual IV values compared to the traditional time series models (VAR and VECM). The VAR and VECM models struggle to capture the time series patterns and shows a lot of fluctuations, which deviate significantly from the actual values, especially for longer forecast horizons.

In contrast, the LSTM and Att-LSTM models demonstrate a better ability to capture the dynamic behavior of the IV time series, including sudden spikes and drops. The Att-LSTM model, in particular, shows the closest alignment with the actual values, suggesting that the attention mechanism helps the model focus on the most relevant information from the historical data.

As the forecast horizon increases, the performance of all models deteriorates, which is expected due to the accumulation of errors and increased uncertainty. However, the deep learning models, especially the Att-LSTM, maintain better performance compared to the VAR and VECM models, even for the 4-hour forecast horizon.

IV Smile Prediction

Figure 8 presents a comparison of the actual and predicted IV smiles for four different models (VAR, VECM, LSTM, and Att-LSTM) at a random time step from the out-of-sample data. The IV smile represents the relationship between the implied volatility and the moneyness of the option contracts. In this figure, the blue lines represent the actual IV smiles, while the orange lines represent the predicted smiles.

The predicted IV smiles from the VAR and VECM models deviate significantly from the actual smile, failing to capture the shape and curvature accurately. These models produce flatter and less pronounced smiles, indicating their limitations in modeling the complex relationships between implied volatility and moneyness.

On the other hand, the LSTM and Att-LSTM models generate IV smiles that closely resemble the actual smile, capturing the overall skew more effectively. The Att-LSTM model, in particular, demonstrates the best performance, producing an IV smile that almost overlaps with the actual smile. This suggests that the attention mechanism in the Att-LSTM model helps capture the intricacies of the IV smile by focusing on the most relevant information from the input data.

IV Surface Prediction

Figure 9 illustrates the predicted implied volatility surfaces by various models (VAR, VECM, LSTM, and Att-LSTM) for a single-step (15 minutes) sliding window forecast for a random contract from the out-of-sample data. The IV surface represents the relationship between implied volatility, moneyness, and time to expiration. The purple surface represents the actual IV surface, while the green surface represents the predicted surface.

The VAR and VECM models produce IV surfaces that deviate significantly from the actual surface, failing to capture the complex dynamics and interrelationships between the three variables. The predicted surfaces from these models appear flatter and less detailed, indicating their limitations in modeling the IV surface accurately.

In contrast, the LSTM and Att-LSTM models generate IV surfaces that closely resemble the actual surface, capturing the overall shape, curvature, and dynamics more effectively. The Att-LSTM model, in particular, demonstrates the best performance, producing an IV surface that exhibits a high degree of similarity to the actual surface. The attention mechanism in the Att-LSTM model helps capture the intricate patterns and dependencies within the IV surface by attending to the most relevant information from the historical data.

The qualitative analysis of the IV time series, smile, and surface predictions reinforces the findings from the quantitative results. The deep learning models, especially the Att-LSTM, consistently outperform the traditional time series models in capturing the complex dynamics and relationships within the implied volatility data. The attention mechanism proves to be a valuable addition, enabling the model to focus on the most relevant information and improve its ability to generate accurate predictions.

5.4 Discussion

The superior performance of deep learning models, particularly the attention-based LSTM (Att-LSTM), in forecasting implied volatility surfaces underscores the importance of capturing complex dynamics and patterns in implied volatility data. The Att-LSTM model's consistent outperformance across various forecasting horizons and strike prices highlights the significance of the attention mechanism in filtering relevant information from historical data, aligning with recent studies that emphasize the benefits of attention in sequence modeling tasks. The study's comprehensive evaluation using multiple metrics and across different forecasting horizons provides a holistic understanding of the models' capabilities. However, the focus on short-term option contracts limits the incorporation of potentially valuable long-term information. One unexpected finding is the slightly better performance of the simpler LSTM model compared to Att-LSTM for the 4-hour forecasting horizon, suggesting potential limitations of the attention mechanism in capturing distant past information. Future research could address this by incorporating longer-term contracts and exploring the impact of data granularity and alternative data sources on predictive power. The findings have significant implications for market participants, including option traders, risk managers, and investors. The superior performance of deep learning models in forecasting implied volatility surfaces can aid in making more informed trading decisions, developing effective hedging strategies, and optimizing portfolio management. The study contributes to the growing literature on machine learning applications in finance, paving the way for further advancements in this field.

5.5 Future Work

The implied volatility surface, reflects how market participants price options across different strikes and expirations. It can vary depending on day of the week and the variation throughout the week can depend on several factors:

Economic Data: Certain days are more prone to significant economic announcements (like employment reports or inflation data), which can spike implied volatility leading up to those events.

Earnings Announcements: If a company is set to report earnings mid-week, options tied to that stock may see increased implied volatility as traders speculate on the outcome.

Time Decay: Options lose time value as expiration approaches. This can affect implied volatility levels throughout the week, often decreasing as the expiry approaches.

Hedge Positions: Institutional traders often adjust their positions at different times during the week, impacting demand for options and, consequently, their implied volatility.

We can address these challenges by developing dynamic models that adapt to temporal patterns in implied volatility based on the specific day of the week, incorporating economic and earnings event calendars as external factors. By integrating news sentiment analysis and data on institutional flows, models could more accurately predict implied volatility shifts around key economic reports and hedge adjustments. This multi-faceted approach could improve the robustness and accuracy of IV surface predictions, ultimately leading to better risk management and trading strategies.

6 Conclusion

This study demonstrates the superior performance of deep learning models, particularly the attention-based LSTM (Att-LSTM), in forecasting implied volatility surfaces for NIFTY50 index options. The Att-LSTM model consistently outperformed traditional econometric models (VAR and VECM) and the standard LSTM model across various forecasting horizons, strike prices, and evaluation metrics. The attention mechanism proved to be a valuable addition, enabling the model to focus on the most relevant information and improve its generalization capability. The comprehensive evaluation using multiple metrics and across different forecasting horizons provided a holistic understanding of the models' capabilities. The deep learning models exhibited a smaller performance gap between in-sample and out-of-sample results compared to the baseline models, suggesting better generalization and less overfitting. The Att-LSTM model's ability to capture complex dynamics and patterns in implied volatility data highlights the importance of advanced machine learning techniques in financial modeling. The findings have significant implications for market participants, including option traders, risk managers, and investors. The superior performance of deep learning models in forecasting implied volatility surfaces can aid in making more informed trading decisions, developing effective hedging strategies, and optimizing portfolio management. The study contributes to the growing literature on machine learning applications in finance, paving the way for further advancements in this field.

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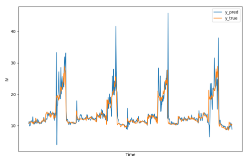
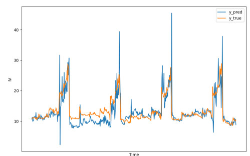


(a) VAR-15min



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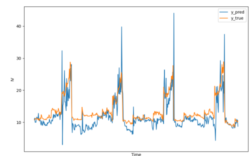

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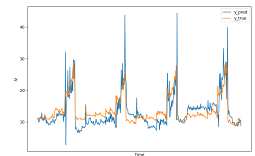
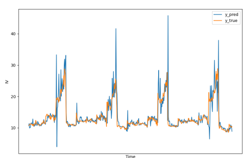


(d) VAR-4H



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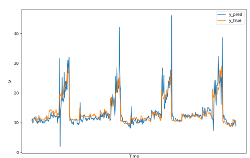
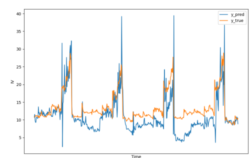
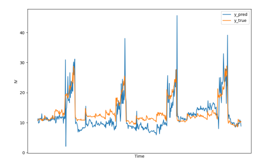


(f) VECM-1H



(g) VECM-2H



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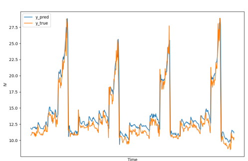

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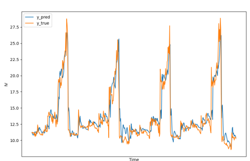

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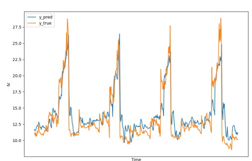
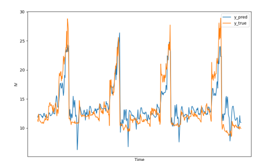
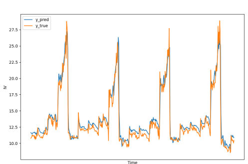


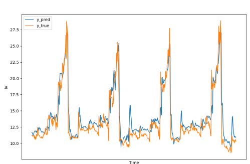
(k) LSTM-2H



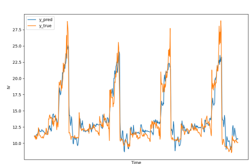
(l) LSTM-4H



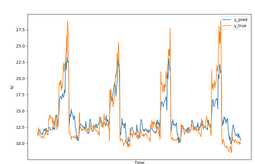
(m) Att-LSTM-15min



(n) Att-LSTM-1H

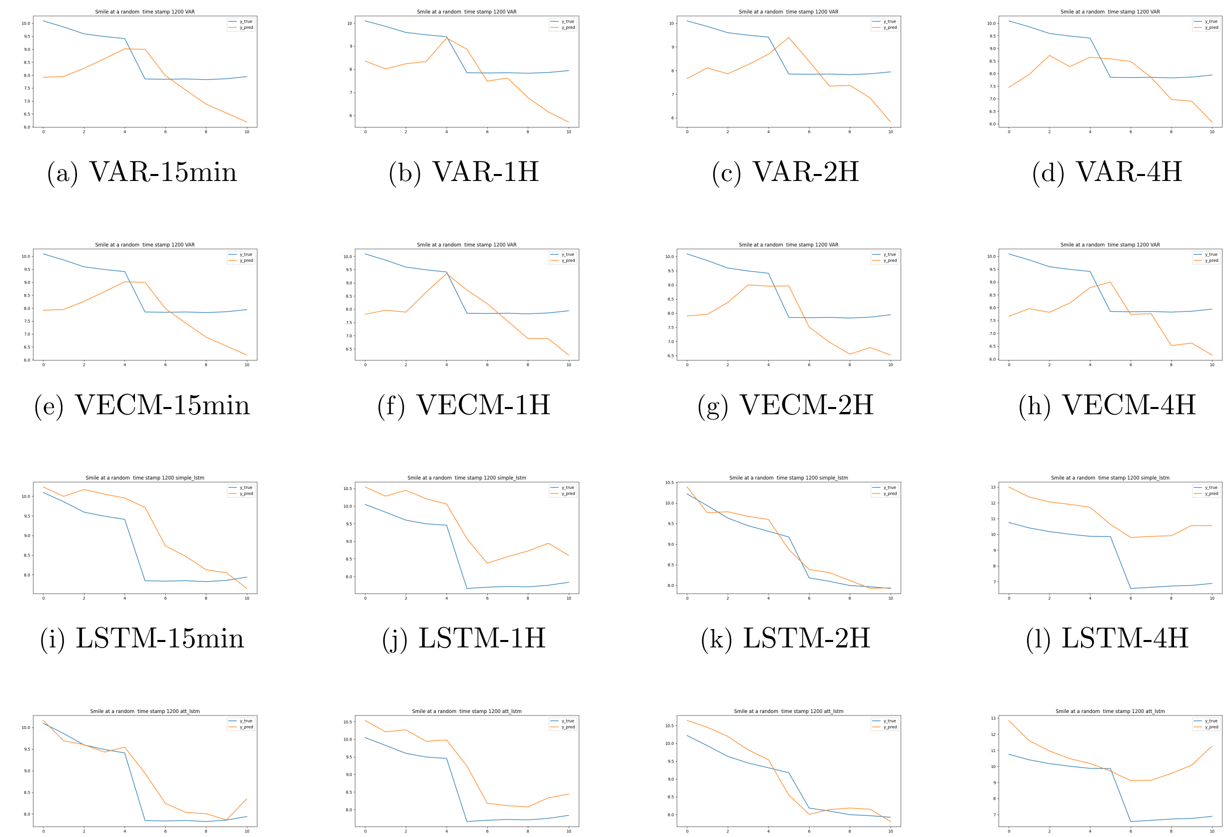


(o) Att-LSTM-2H



(p) Att-LSTM-4H

Figure 7: Comparison of actual and predicted implied volatility (IV) time series for the ATM contract using four different models: VAR, VECM, LSTM, and Att-LSTM. The predictions are made on the out-of-sample data over four forecast horizons: 15 minutes, 1 hour, 2 hours, and 4 hours.



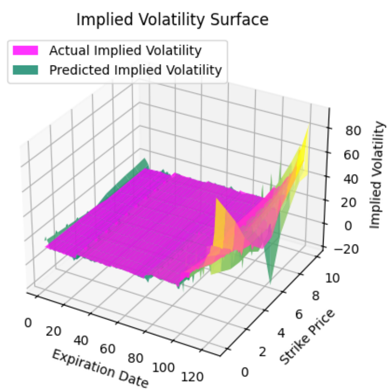
(a) VAR-15min (b) VAR-1H (c) VAR-2H (d) VAR-4H

(e) VECM-15min (f) VECM-1H (g) VECM-2H (h) VECM-4H

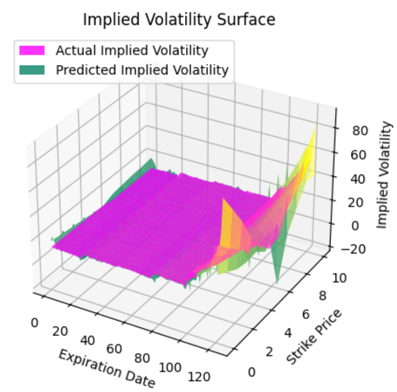
(i) LSTM-15min (j) LSTM-1H (k) LSTM-2H (l) LSTM-4H

(m) Att-LSTM-15min (n) Att-LSTM-1H (o) Att-LSTM-2H (p) Att-LSTM-4H

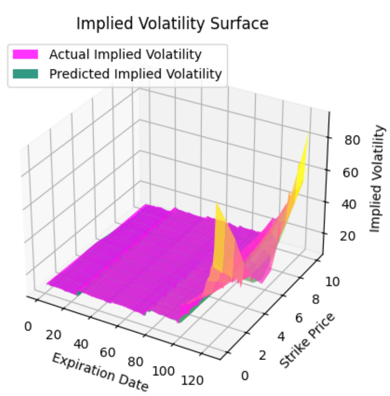
Figure 8: Comparison of actual and predicted IV smiles of four different models: VAR, VECM, LSTM, and Att-LSTM. The predictions are made over four forecast horizons: 15 minutes, 1 hour, 2 hours, and 4 hours at a random time-step from the out-of-sample data



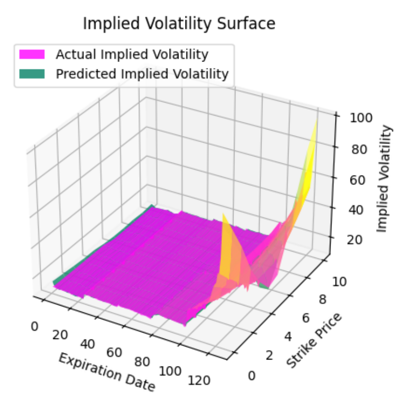
(a) VAR surface



(b) VECM surface



(c) LSTM Surface



(d) Att-LSTM Surface

Figure 9: Predicted Implied Volatility Surfaces by various models for single step (15 mins) sliding window forecast for a random contract from out-of-sample data.